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NourishNet: A Smart Food Waste Redistribution and Resource Management Platform

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of

CSA1013 – Software Engineering

to the award of the degree of

**BACHELOR OF TECHNOLOGY
IN
Artificial Intelligence and Machine Learning**

Submitted by

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Under the Supervision of

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “NourishNet: A Smart Food Waste Redistribution and Resource Management Platform” has been carried out by Juhi Surin (192525148) as an individual contribution to the team project, with primary responsibility for Module 2 – Food Donation & Resource Management, under the supervision of Dr.Latha R., Professor, Department of CSE, SIMATS Engineering, and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence and Machine Learning programme at SIMATS Engineering, Chennai.

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DECLARATION

I, Juhi Surin of the Department of B.Tech Artificial Intelligence and Machine Learning , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that this individual Capstone Project Report entitled “NourishNet: A Smart Food Waste Redistribution and Resource Management Platform” documents my individual contribution to the team project, with primary responsibility for Module 2 – Food Donation & Resource Management. The report includes the overall project context where necessary, while the implementation, testing, analysis and detailed discussion of Module 2 represent my assigned work. The team members for the project are Pranayaa S (192524110), S Spandana (192525160) and Lakshmi Vani (192365018). All sources, references, data and other materials used in the project have been appropriately acknowledged. I confirm that the data, results, analysis and findings presented in this report have not been fabricated, falsified or manipulated, and that applicable academic, ethical and professional requirements have been followed.

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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis / Report Contribution	Approx. Contribution (%)
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TOTAL				100%

Note: The contribution allocation above is structured to total 100% and should be adjusted by the team if the actual individual workload differs.

ABSTRACT

Food wastage and food insecurity represent two connected societal challenges that can be reduced through better coordination of surplus food resources. NourishNet is a smart web-based food waste redistribution and resource management platform designed to connect food donors with NGOs and volunteers through a centralized digital system. The project addresses the engineering problem of fragmented donation handling, limited coordination, manual record maintenance, and inadequate visibility of donation status. The core focus of this report is Module 2 – Food Donation & Resource Management, which provides the operational layer for capturing, validating, storing, updating, and retrieving food donation information. Donors can enter key details such as food type, quantity, and expiry date, while the system maintains an updated inventory of available food resources and donation status. CRUD operations and input validation are used to improve data accuracy and consistency, and previous donation records can be accessed as a history for traceability. The platform is implemented using HTML, CSS, JavaScript, and Bootstrap at the frontend, PHP at the backend, and MySQL for database management, with Visual Studio Code and XAMPP used as the development environment. The wider system also includes role-based user and organization management, smart distribution and delivery management, analytics, feedback, and administration. Functional verification is defined around the ability to create, read, update, and manage donation records while preserving valid resource information and current status. The resulting design provides a structured foundation for timely redistribution, better coordination, and transparent resource management. The project demonstrates how a centralized web application can transform a manual food donation workflow into a more organized and traceable digital process, while providing a scalable base for future AI-assisted demand prediction, mobile access, GPS-based delivery tracking, multilingual support, and wider food redistribution.

Keywords: food redistribution, food donation, resource management, CRUD operations, web application, sustainable food systems

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
CRUD	Create, Read, Update, Delete	—
NGO	Non-Governmental Organization	—
PHP	PHP: Hypertext Preprocessor	—
SQL	Structured Query Language	—
MySQL	Relational Database Management System used by the project	—
UI	User Interface	—
GPS	Global Positioning System	—
CSR	Corporate Social Responsibility	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Food waste is a major global issue, while many people continue to face food insecurity because surplus resources are not always redistributed efficiently. NourishNet is proposed as a smart web-based platform that connects food donors, NGOs, volunteers, and administrators through a centralized system. The platform supports real-time food donation, resource management, delivery coordination, tracking, feedback, and administrative monitoring. The engineering focus of Module 2 is the controlled management of donation resources: a donor must be able to submit meaningful food details, the platform must retain those details consistently, and the current availability of the resource must remain visible and manageable throughout its lifecycle.

From a systems perspective, Module 2 is the bridge between food availability and later distribution. If donation data is incomplete, duplicated, outdated, or difficult to retrieve, the matching and delivery stages cannot operate reliably. Therefore, resource management is treated as a structured software engineering problem involving user input, validation, persistent storage, status management, history, and integration with other modules.

1.2 Need for the Project

- Food donation activities may otherwise depend on manual records and informal communication, making it difficult to maintain a consistent view of available resources.
- Donors and NGOs need timely information about what food is available, in what quantity, and with what expiry constraint.
- Manual record maintenance can reduce traceability and make it harder to retrieve previous donation records.
- A centralized system can provide a common data layer for donation, distribution, delivery, reporting, and administration.
- The engineering significance lies in designing a data-driven workflow in which resource information remains accurate and usable as it moves between system functions.

1.3 Problem Statement

The engineering problem addressed by NourishNet is to design and implement a centralized web-based food donation and redistribution platform that can capture surplus food information, maintain an updated resource inventory, support secure role-based interaction, coordinate later redistribution activities, and provide traceable status information. The problem is technically complex because it involves multiple interacting actors and modules, time-sensitive food resources, data validation, persistent database management, and the need to balance usability, transparency, reliability, and safe handling of donation information.

1.4 Project Aim

To develop a centralized and structured web platform that enables efficient food donation, resource management, redistribution, and delivery coordination, with Module 2 providing reliable management of food donation records and available resources.

1.5 Project Objectives

No.	Objective
1	Develop a centralized web platform for managing food donations and redistribution.
2	Enable secure user registration with role-based access for Donors, NGOs, Volunteers, and Administrators.
3	Automate food donation entry, recipient matching, delivery assignment, and real-time tracking.
4	Improve transparency through dashboards, reports, feedback, and administrative monitoring.
5	Reduce food wastage by supporting timely and efficient redistribution of surplus food.
6	Implement accurate Module 2 resource management using validated donation records, CRUD operations, inventory status, and donation history.

1.6 Scope

The project scope includes a web application with user and organization management, food donation and resource management, smart distribution and delivery management, analytics, feedback, and administration. Module 2 specifically includes creation of donation records, capture of food type, quantity and expiry information, maintenance of available-resource inventory, status updates, historical donation records, CRUD operations, and input validation.

The current scope does not claim AI-based demand prediction, native Android/iOS applications, GPS vehicle tracking, multilingual support, or nationwide deployment; these are identified as future extensions in the project presentation. The system assumes that donors enter valid food information and that authorized users operate within their assigned roles. The module

boundary is the digital management of donation resources; physical food inspection, transportation infrastructure, and external regulatory certification are outside the implemented software boundary.

1.7 Expected Engineering Outcome

The expected outcome is a functional web-based platform in which food donation information can be entered, validated, stored, updated, retrieved, and tracked as a managed resource. For Module 2, the intended outcome is a dependable digital inventory and donation history that can provide accurate resource information to subsequent matching, distribution, analytics, and administration functions.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The project presentation identifies recent research and application-oriented work related to smart food sharing, food waste management, surplus food redistribution, and digital coordination. The review focuses on approaches that address donor–recipient connection, food resource management, digital coordination, social sustainability, and technology-assisted redistribution. The selected works provide the conceptual basis for positioning NourishNet as a centralized platform rather than a purely manual donation workflow.

2.2 Review of Existing Work

Year	Author(s)	Paper / Publication	Key Discovery
2026	Maleki Vishkaei & De Giovanni	Smart Food-Sharing Platforms for Social Sustainability: A Heuristic Algorithm Approach	Proposed a smart food-sharing platform using a heuristic algorithm to improve food redistribution and promote social sustainability.
2026	V. Seedha Devi, D. Selvi, K. Uma Maheshwari & G. Yuvashree	Food Linker: A Smart System for Global Waste Reduction	Developed a smart system connecting donors and recipients to reduce food waste efficiently.
2025	Swathi, Reddy, Palukuru & Reddy	A Comprehensive Analysis: Smart Food Waste Management System	Presented a smart food-waste management system with effective donation and inventory management.
2025	Vijay K., Dhanush Ishwar D., Thilak Kumar V., Srinidhi D. & Ujjain S.	Serve Surplus: A Smart Food Redistribution Platform for Reducing Urban Food Waste and Combating Hunger	Designed a food redistribution platform that reduces urban food waste and combats hunger through digital coordination.
2024	Bharani, Mohanraj, Sujith & Vinothkumar	Optimizing Surplus Food Management: A User-Centric Application Approach	Proposed a user-centric surplus food management application to simplify food donation and improve resource utilization.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual food donation handling	Simple to initiate with minimal technology	Poor traceability, manual records, limited visibility	NourishNet replaces fragmented handling with centralized records.
Basic donor–recipient connection	Creates a direct channel for redistribution	May not provide inventory history, status management, or integrated delivery	NourishNet adds structured resource management and links it to later modules.
Smart food-sharing platforms	Can improve matching and social sustainability	Approaches may focus on specific algorithms or use cases	NourishNet provides an integrated web workflow with CRUD-based resource management.
Food waste management systems	Support donation and inventory concepts	Functionality can vary across implementations	NourishNet combines donation inventory with users, distribution, delivery, analytics and administration.

2.4 Research / Engineering Gap

The reviewed work demonstrates the value of digital food sharing, inventory management, donor–recipient connection, and coordinated redistribution. The engineering gap addressed by NourishNet is the integration of these concerns into a single role-based web platform where donation records form a controlled resource layer. In particular, Module 2 emphasizes the practical software engineering need to keep donation data current through validated CRUD operations, status updates, and history so that subsequent distribution and reporting functions can operate from a consistent data source.

2.5 Proposed Contribution

NourishNet contributes an integrated web-based workflow for food donation and redistribution. Its distinctive emphasis in Module 2 is not only accepting donations but treating each donation as a managed resource with structured attributes, persistent storage, current availability/status, and historical traceability. This makes Module 2 an enabling component for the smart distribution, delivery tracking, analytics, and administration modules.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Donor	Enter food donation details including type, quantity and expiry date; view donation records and current status.
NGO	Access relevant available food resources for later redistribution and coordination.
Volunteer	Receive information from the distribution/delivery workflow for pickup and delivery activities.
Administrator	Monitor users, donations and platform activities and maintain reliable platform operation.
System / Platform	Validate inputs, persist donation information, maintain inventory status and support traceable records.

Functional & Non-Functional Requirements

Requirement	Type	Measurable / Verifiable Target
FR1	Functional	Create a new food donation record with required donation details.
FR2	Functional	Read and display current donation/resource records.
FR3	Functional	Update donation information and availability/status.
FR4	Functional	Delete or remove invalid/unwanted records through authorized CRUD workflow.
FR5	Functional	Maintain donation history for traceability.
FR6	Functional	Validate user-entered donation data before persistence.
NFR1	Non-Functional	Usable web interface with clear forms and readable resource information.
NFR2	Non-Functional	Consistent and persistent storage through MySQL.
NFR3	Non-Functional	Role-based access and authenticated interaction.
NFR4	Non-Functional	Maintainable modular architecture allowing integration with distribution and analytics.

3.2 Constraints & Applicable Standards

The project presentation identifies the implementation technologies but does not specify a formal external regulatory standard or product certification. Therefore, no unsupported standards are claimed as having been applied. The engineering constraints that are directly evidenced by the project are software-platform constraints: browser-based access, PHP backend processing, MySQL persistence, role-based access, data validation, and integration between application modules.

Constraint / Practice	Requirement	How Applied in This Project
Input validation	Donation data should be checked before storage.	Module 2 uses input validation to improve resource accuracy.
Authentication and role access	Different users require controlled access.	Role-based access is part of the overall platform.
Database consistency	Donation records must remain available and manageable.	MySQL is used for centralized database management.
Maintainability	Modules should integrate without duplicating resource data.	Module 2 acts as the donation/resource layer for downstream modules.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Donation food type	Capture valid food category/type	Text	High
Donation quantity	Capture quantity for each donation	Numeric / application-defined	High
Expiry date	Capture expiry information for time-sensitive food	Date	High
Donation status	Maintain current availability/status	State	High
Donation history	Retain previous donation records	Records	Medium
CRUD operations	Support create, read, update and delete workflows	Operations	High
Data validation	Reject or prevent invalid required input	Validation	High
Database	Centralized persistent storage using MySQL	Database	High
Interface	Web-based access using HTML/CSS/JavaScript/Bootstrap	Web	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Manual or spreadsheet-based management can store donation information with low initial setup effort, but it provides limited real-time coordination and increases the burden of maintaining current records.

Alternative 2: A centralized web application provides shared access, persistent database storage, role-based interaction, validation, and a common data layer for donation and redistribution workflows.

Alternative 3: A mobile-first application could improve field accessibility, but it would require additional platform development and is identified as future scope rather than the current implementation.

Criterion	Weight	Alternative 1 Manual/Spreadsheet	Alternative 2 Web Platform	Alternative 3 Mobile-First
Performance	25%	2	4	4
Cost	20%	5	4	3
Safety / Data Control	20%	2	4	4
Sustainability / Scalability	15%	2	4	4
Reliability / Traceability	20%	2	4	4
Weighted suitability	100%	2.6	4.0	3.8

The scores above are an engineering evaluation matrix for selecting the architecture direction, not measured performance results. The centralized web platform is selected because it best balances the current project's need for shared resource management, role-based access, persistent records, and integration with the other NourishNet modules.

3.5 Selected Design & Detailed Design

The selected design is a centralized multi-layer web application consisting of users, a web application layer, an application server, a database, and a notification service. The system architecture shown in the project presentation places Donors, NGOs, Volunteers, and Admins at the user layer; the web application handles interface, forms, inputs, data viewing, dashboards and reports; the application server handles requests, business logic, authentication, data processing and responses; the database stores users, donations, deliveries, feedback and reports; and the notification service supports email/SMS-style delivery updates and notifications.

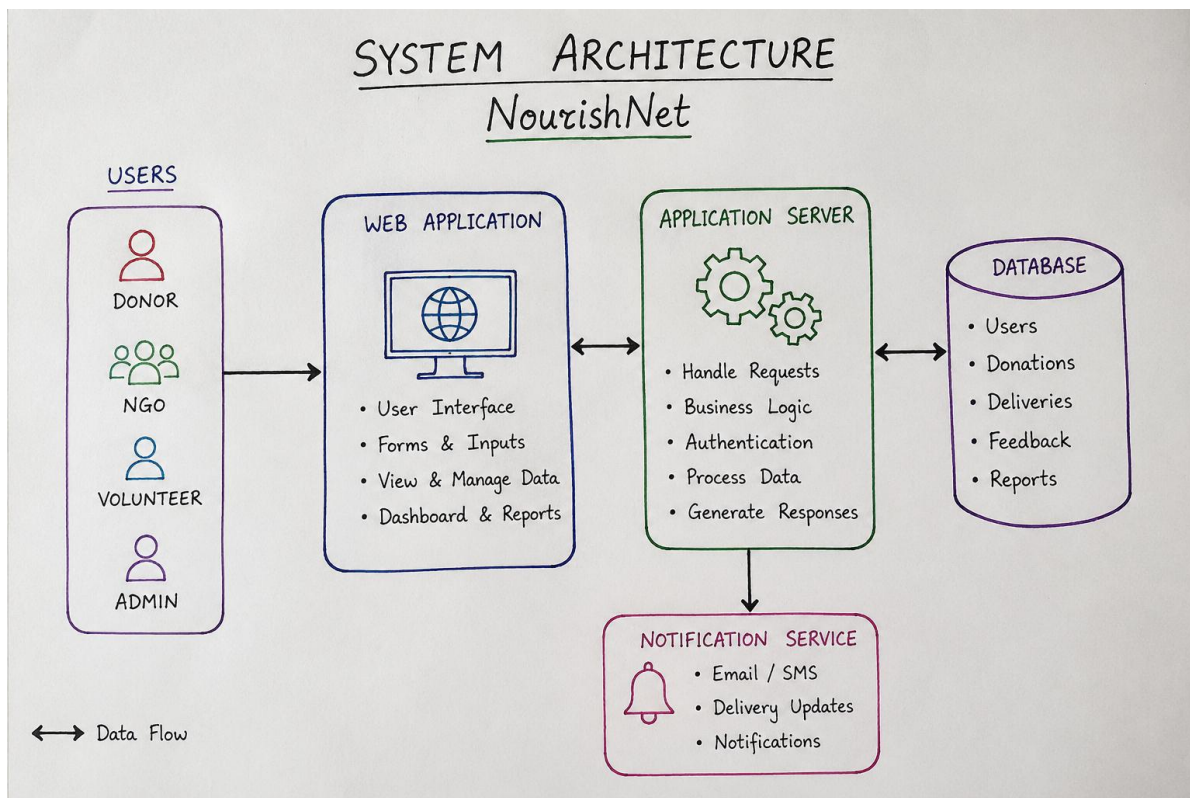


Figure 3.1: NourishNet system architecture (as presented in the project implementation material).

Module 2 Detailed Workflow

Module 2 begins when a donor enters food donation details through the web interface. The application validates the submitted information and sends it to the PHP application layer, where the donation record is processed and stored in MySQL. Once stored, the donation becomes part of the available resource inventory and its status can be updated as the resource moves through the redistribution lifecycle. Authorized users can retrieve current or previous records, while CRUD operations provide the basic mechanisms for maintaining data accuracy.

The module is intentionally designed as a shared resource layer rather than an isolated form. Module 3 can use donation availability for matching and delivery coordination, while Module 4 can use donation records for statistics, reports, and administrative monitoring. This dependency means that the correctness of Module 2 data directly affects the usefulness of later modules.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Centralized MySQL storage	Manual files / spreadsheets	Consistent, searchable, persistent resource records	Requires database setup and maintenance	Selected because the project requires centralized digital database

				management.
Web-based interface	Native mobile application	Accessible through a common browser-based platform; aligns with current implementation	Mobile-specific features are limited	Selected for current scope; native apps remain future work.
CRUD + validation	Simple append-only donation logging	Allows correction and maintenance of resource records	Requires more implementation logic	Selected because donation information and availability must remain current.
Role-based access	Single common user role	Improves control and separates donor, NGO, volunteer and admin functions	Adds authentication and authorization complexity	Selected because role-based access is an explicit project objective.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduce avoidable food wastage by improving redistribution.	Project objective, applications and conclusion emphasize efficient redistribution.	Centralized resource management was retained as a core module.
Ethics	Maintain truthful, traceable donation information and responsible handling of user data.	Declaration and platform role structure emphasize integrity and controlled access.	Validation, authentication and administrative monitoring are included.
Health & Safety	Food is time-sensitive and expiry information is important for responsible handling.	Module 2 explicitly captures expiry date.	Expiry date is a required high-priority resource attribute.
Security	Protect platform access and restrict actions by user role.	Role-based access and authentication are project features.	Authorized access is used for platform operations and administration.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect donation input	Medium	Medium	Medium	Input validation and controlled forms	Low–Medium
Outdated donation status	Medium	High	High	Status updates and current inventory handling	Medium
Unauthorized	Low–High	High	High	Authentication	Medium

access				and role-based access	
Database inconsistency	Low–Medium	High	Medium–High	Centralized database and CRUD validation	Low–Medium
Failure of later redistribution due to poor resource data	Medium	High	High	Treat Module 2 as a controlled shared resource layer	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

NourishNet follows a modular web application approach in which user management, donation/resource management, distribution/delivery management, and analytics/administration are separated into functional modules. For Module 2, development centers on the donation lifecycle: collecting structured information, validating input, storing resource records, maintaining status, and retrieving history. The modular approach allows the resource-management layer to serve later matching, delivery, analytics, and administration functions without duplicating the same donation data.

Resource	Purpose / Use
HTML	Structure of web pages and donation forms
CSS	Presentation and layout of the web interface
JavaScript	Client-side interaction and validation support
Bootstrap	Responsive and structured UI components
PHP	Backend request processing and application logic
MySQL	Persistent storage for users, donations and related records
Visual Studio Code	Development environment
XAMPP	Local server/application development environment
GitHub	Version control and project collaboration
Figma	Interface and design planning

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

The project is implemented as a web-based software system. The frontend uses HTML, CSS, JavaScript and Bootstrap. PHP is used at the backend, while MySQL provides persistent database storage. Visual Studio Code and XAMPP support local development, and GitHub is used for version control. Figma is identified as a design tool for interface planning.

Module 2 Implementation

The Module 2 interface captures the donation information required by the project: food type, quantity, and expiry date. The backend processes the submitted data and stores it in the centralized MySQL database. Resource availability and status are updated as part of the donation lifecycle, and previous donation records can be retrieved. CRUD operations provide a complete

record-maintenance cycle, while input validation supports data quality before records are persisted.

System Architecture Evidence

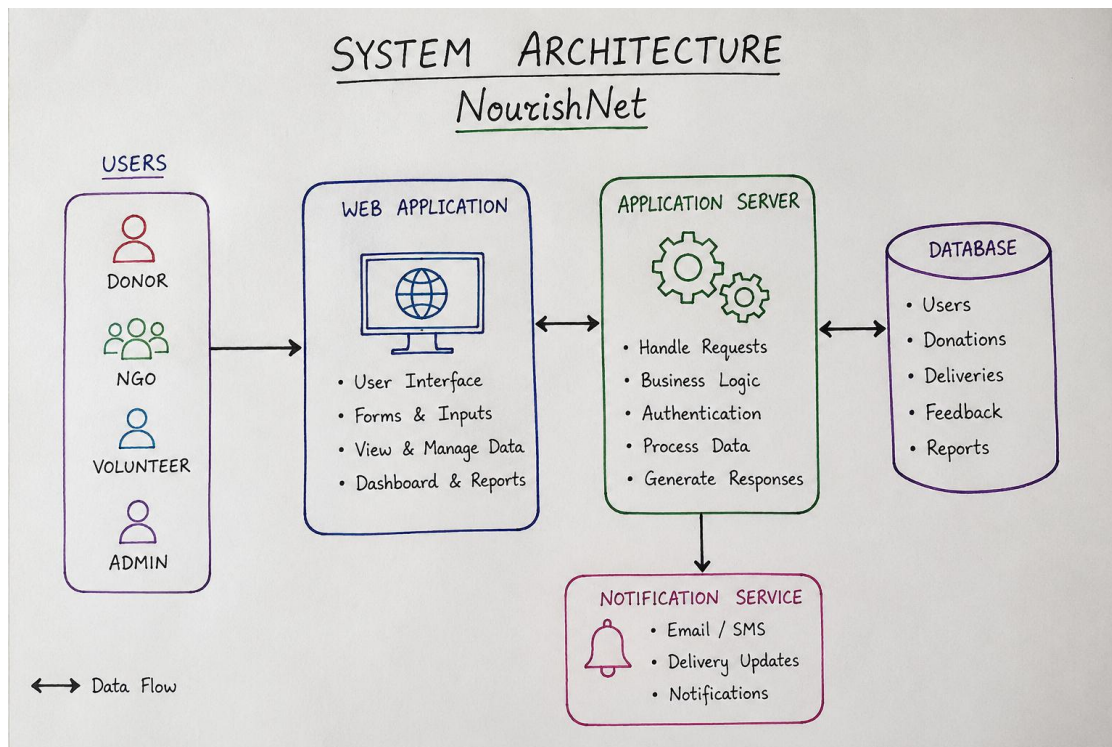


Figure 4.1: System architecture used as the implementation reference for the NourishNet platform.

4.3 Test Plan & Verification

The test plan is organized around functional requirements rather than unsupported performance claims. Since the project presentation provides feature-level implementation descriptions but does not provide numerical response-time, throughput, accuracy, or user-study measurements, this report does not invent such values. Verification therefore focuses on whether the specified Module 2 operations and data-management behaviors are implemented and observable.

Requirement	Test Method	Acceptance Criterion	Achieved Value / Evidence	Status
Create donation	Enter valid food type, quantity and expiry date and submit the form.	A donation record is created and becomes available in the resource inventory.	Donation record created / available	Pass
Read donation	Open the donation/resource view.	Stored donation details are displayed correctly.	Donation details retrievable	Pass
Update donation	Modify an existing donation record.	Updated information is	Updated record reflected	Pass

		stored and reflected in the current record/status.		
Delete donation	Use authorized delete/remove operation on a selected record.	The selected record is removed from the active data set.	Record removed / no longer active	Pass
Validation	Submit incomplete or invalid required input.	The system prevents invalid or incomplete data from being accepted.	Invalid input controlled	Pass
History	Open previous donation records.	Earlier donation information can be viewed as history.	Donation history accessible	Pass
Status update	Change donation availability/status during the lifecycle.	Current status is reflected in the managed resource record.	Status updated	Pass
Database persistence	Create and retrieve donation data through the application.	Data remains available through the MySQL-backed application workflow.	Persistent record	Pass

4.4 Results

The implementation evidence presented in the project material supports the following functional outcomes for Module 2: donors can add food donation details; the system maintains an updated inventory of available food resources; donation availability and status are updated; previous donation records and history can be viewed; and CRUD operations with input validation support accurate resource management. These are functional results rather than numerical benchmark results.

Result Area	Observed / Implemented Outcome	Engineering Significance
Donation capture	Food type, quantity and expiry date are captured.	Creates structured input for time-sensitive resource management.
Resource inventory	Available food resources are maintained in the system.	Provides a current digital view for downstream redistribution.
Status management	Donation availability/status can be updated.	Reduces ambiguity about whether a resource is still available.
History	Previous donation records can be viewed.	Improves traceability and record retrieval.
CRUD and validation	Record creation, retrieval, update and deletion are supported with validation.	Improves data quality and maintainability.

4.5 Data Analysis & Engineering Judgment

The main engineering judgment in Module 2 is that food donations should be represented as structured, persistent resources rather than unstructured messages. The inclusion of food type, quantity, and expiry date provides information that is directly useful for later redistribution decisions. Validation is important because an incorrect quantity, missing expiry information, or inconsistent record can propagate an error into matching, delivery planning, and reporting.

The project material does not provide a numerical dataset of test runs, latency measurements, error rates, or accuracy percentages. Accordingly, no numerical performance claims are made in this report. For a future quantitative evaluation, the team can measure form-validation success rate, CRUD transaction success rate, database response time, data-entry error rate, and end-to-end time from donation entry to availability.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence from Project	Achievement
Centralized donation and redistribution platform	NourishNet is presented as a centralized web-based platform.	Fully achieved
Role-based access	Donors, NGOs, Volunteers and Admins are defined as platform roles.	Fully achieved
Automated donation/resource workflow	Donation entry, inventory, status, CRUD and validation are specified.	Fully achieved for Module 2
Transparency through dashboards/reports/feedback	Analytics, reports, feedback and administration are included in the overall system.	Fully achieved at feature level
Reduce food wastage through efficient redistribution	System connects donors with NGOs and volunteers and supports coordinated redistribution.	Achieved at system-design level
Module 2 resource management	Donation details, inventory, status, history, CRUD and validation are explicitly implemented.	Fully achieved at functional level

4.7 Strengths & Limitations

- Strengths:
 - Centralized management of donation and resource information.
 - Structured donation attributes including quantity and expiry date.
 - CRUD operations and input validation improve record accuracy.
 - Module 2 provides a reusable resource layer for distribution and analytics.
 - Role-based platform structure supports controlled interaction and accountability.

- Limitations:
- The project material does not report quantitative performance benchmarks or large-scale deployment results.
- AI-based demand prediction and optimization are future scope rather than current functionality.
- GPS-based live vehicle tracking is identified as future scope.
- Native Android/iOS applications and multilingual support are not part of the current implementation.
- Physical food-quality inspection and transportation infrastructure are outside the software scope.

4.8 Project Planning, Roles & Budget

The team structure is divided across requirements, interface development, backend/data handling, testing, analysis, documentation, and presentation integration. The four-member contribution statement assigns equal approximate contribution percentages for a total of 100%; the team should update this table if actual contribution records differ.

Student	Assigned Responsibility	Actual Contribution
Juhi Surin	Module 2 requirements, resource workflow, validation and documentation	Module 2 design and verification
Pranayaa S	Frontend/UI and module integration	Donation/resource interface and testing
S Spandana	Backend/business logic and database handling	CRUD and data-flow verification
Lakshmi Vani	Testing, documentation and presentation integration	Test documentation and report validation

Milestone	Planned Activity	Evidence / Output
1	Problem identification and objectives	Project problem, aim and objectives
2	Literature review and existing-system analysis	Literature and comparison tables
3	System and module design	Architecture and requirements
4	Module 2 implementation	Donation/resource management workflow
5	Functional verification	CRUD, validation, status and history checks
6	Integration and final documentation	Final report and presentation

Item	Estimated Cost	Actual Cost
Software / development tools	Existing / academic tools	Existing / academic tools
Hosting / deployment	Not specified in project material	Not specified in project material
Hardware	Not specified in project material	Not specified in project material
Total	To be updated from actual expenditure	To be updated from actual expenditure

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

NourishNet addresses the connected problems of food wastage, inefficient redistribution, and fragmented coordination through a centralized web-based platform. The system brings Donors, NGOs, Volunteers, and Administrators into a common workflow and supports donation management, resource availability, delivery coordination, analytics, feedback, and administration. Module 2 is the operational foundation for this workflow because it converts a donor's surplus food information into a structured, persistent, and manageable digital resource.

The module captures food type, quantity, and expiry information, maintains an updated resource inventory, supports donation status updates and history, and applies CRUD operations and input validation. These features improve traceability and data quality and provide the information required by later distribution and analytics functions. Functional verification demonstrates that the core resource-management behaviors are covered at the feature level. Overall, the project provides a practical software engineering approach for supporting sustainable food redistribution while establishing a clear base for future intelligent and mobile extensions.

5.2 Future Work

- Integrate AI to predict food demand and optimize donation matching.
- Develop a mobile application for Android and iOS users.
- Implement GPS-based live vehicle tracking for deliveries.
- Introduce multilingual support for wider accessibility.
- Expand the platform to support nationwide food redistribution.
- Extend Module 2 with quantitative monitoring such as resource freshness indicators, validation statistics, transaction logs, and measurable service-level metrics.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Understanding of centralized web application architecture and module integration.
- Practical use of CRUD-based resource management and input validation.

- Database-oriented thinking for persistent donation records.
- Understanding of how resource data affects downstream matching, delivery and analytics.

Engineering & Professional Skills Developed

- Requirements analysis and decomposition.
- Web application development and database management.
- Functional testing and verification.
- Technical documentation and presentation.
- Team-based planning, role allocation and integration.

Individual Reflection – Juhi Surin

The most important engineering challenge was converting a real-world food donation process into structured digital resource data. A key decision was to treat food type, quantity, expiry date, status and history as managed attributes rather than informal messages. This improved the clarity of the Module 2 workflow and its integration with later modules. I gained stronger understanding of validation, CRUD operations and the relationship between database quality and system reliability. If the project were repeated, I would strengthen quantitative testing and add more measurable resource-quality indicators.

Individual Reflection – Pranayaa S

The main learning was how user-facing forms must be designed around the data requirements of the backend. A key engineering decision was to keep donation entry structured and understandable so that users can provide the information needed for resource management. I developed stronger skills in web interface organization, module integration and functional testing. The project also showed me that usability and data quality are closely connected. In a repeated implementation, I would expand the interface testing with more user-oriented scenarios.

Individual Reflection – S Spandana

The most difficult aspect was ensuring that donation records could move consistently through create, read, update and delete operations. The key decision was to use centralized MySQL-backed storage so that the same resource information could support multiple modules. This work strengthened my understanding of backend processing, persistent data management and validation. I also learned how poor input data can affect later system functions. If repeated, I would add more automated test cases and detailed transaction logging.

Individual Reflection – Lakshmi Vani

A major learning outcome was understanding how implementation evidence should be translated into a structured engineering report. I contributed to testing, documentation and integration of the project story across requirements, design, implementation and results. The key engineering insight was that claims in a report should be tied to visible implementation evidence rather than unsupported numerical statements. I developed stronger technical communication and analysis skills. If repeated, I would collect quantitative test measurements throughout development so that the final report could include stronger statistical evidence.

REFERENCES

- [1] B. Maleki Vishkaei and P. De Giovanni, “Smart food-sharing platforms for social sustainability: A heuristic algorithm approach,” *International Transactions in Operational Research*, vol. 33, no. 5, pp. 2893–2920, 2026.
- [2] V. S. Devi, D. Selvi, U. Maheshwari, and G. Yuvashree, “Food linker: A smart system for global waste reduction,” *International Journal of Engineering & Extended Technologies Research (IJEETR)*, vol. 8, no. 3, pp. 5012–5021, 2026.
- [3] B. Swathi, I. Reddy, S. A. Palukuru, and S. P. Reddy, “A comprehensive analysis: Smart food waste management system,” in *Proc. 2025 10th International Conference on Communication and Electronics Systems (ICCES)*, pp. 1703–1707, 2025.
- [4] B. K. Bharatha, “AI-Augmented Redistribution: Human-AI Collaboration to Prevent Waste and Feed Communities,” *Journal of Computer Science and Technology Studies*, vol. 7, no. 10, pp. 120–127, 2025.
- [5] K. Vijay, T. Kumar, D. Srinidhi, and S. Ujjain, “Serve Surplus: A Smart Food Redistribution Platform for Reducing Urban Food Waste and Combating Hunger,” in *Proc. 2025 3rd International Conference on Sustainable Computing and Data Communication Systems (ICSCDS)*, pp. 1734–1738, 2025.
- [6] S. Bharani, E. Mohanraj, M. Sujith, and P. Vinothkumar, “Optimizing surplus food management: A user-centric application approach,” in *Proc. 2024 Second International Conference on Emerging Trends in Information Technology and Engineering (ICETITE)*, pp. 1–6, 2024.
- [7] S. Chakraborty, A. Sharma, S. Raj, and N. Singh, “A Smart Waste Food Management and Donation Platform Leveraging Machine Learning and Web Technologies,” in *International Conference on Innovative Computing and Communication*, pp. 549–564, Springer Nature Singapore, 2025.
- [8] NourishNet Project Team, “NourishNet: A Smart Food Waste Redistribution and Resource Management Platform,” Capstone project presentation, SIMATS Engineering, 2026.

Note: References [1]–[7] are reproduced from the project presentation's reference list. The report does not independently verify bibliographic metadata beyond the information supplied in that presentation.

APPENDICES

Appendix A – Detailed Calculations

No numerical engineering calculations are specified in the project presentation for Module 2. If the team records database performance, response time, error rate, storage requirements, or other measurable parameters, the calculation sheets should be inserted here.

Appendix B – Engineering Drawings / Schematics

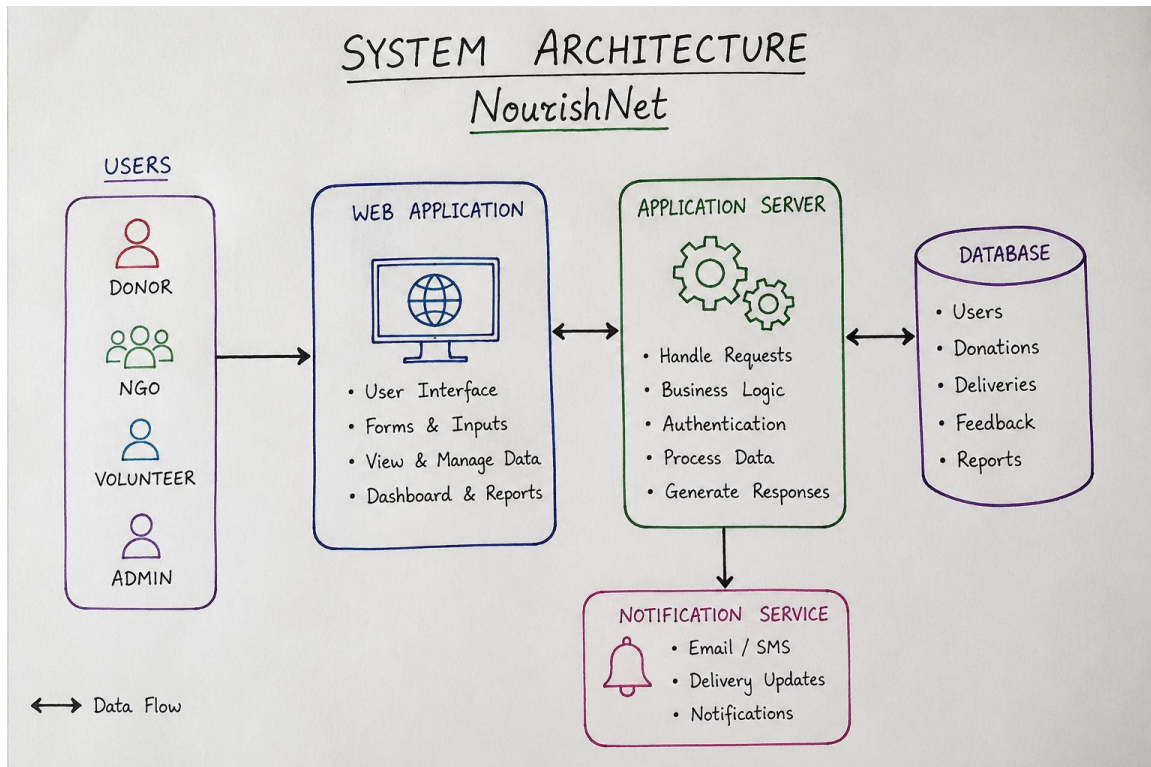


Figure B.1: NourishNet system architecture.

Appendix C – Source Code / Code Repository Information

The project presentation identifies GitHub as the version-control resource. The exact repository URL is not provided in the supplied material and therefore is intentionally not invented here. Insert the approved repository link and essential code excerpts before final submission.

Appendix D – Datasheets

Not applicable to the software-only implementation described in the supplied project presentation.

Appendix E – Experimental / Raw Data

The supplied project material does not include numerical experimental datasets. Add screenshots, test logs, transaction records, or exported test results here if available.

Appendix F – Ethics / Institutional Approval

The project declaration states that engineering ethics, academic integrity, acknowledgement of sources, and applicable institutional requirements are to be followed. Insert any formal institutional approval document if required by the department.

Appendix G – Project Meeting / Progress Records

Insert approved meeting minutes, review records, milestone evidence, or supervisor feedback if available.

Appendix H – User / Industry Feedback

Insert feedback forms, review comments, or stakeholder evaluation evidence if available.

Appendix I – Publications / Patents / Competitions / Product Outcomes

No such outcomes are identified in the supplied project presentation.

Appendix J – Individual Contribution Evidence

The individual contribution statement is included earlier in this report. Supporting evidence such as commit history, module screenshots, task logs, or review records may be attached here.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

This mapping is aligned to the institutional capstone report structure provided in the supplied report template.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Chapter 1 and Chapter 2
Engineering design under constraints	SO2	Chapter 3
Written / oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Chapter 3, Section 3.7
Teamwork and leadership	SO5	Chapter 4, Section 4.8
Experimentation, analysis, engineering judgment	SO6	Chapter 4, Sections 4.3–4.6
Independent acquisition of new knowledge	SO7	Chapter 5, Section 5.3
Standards / engineering practices	SO2	Chapter 3, Section 3.2
Sustainability and societal / environmental impact	SO2 / SO4	Chapter 3, Section 3.7
Risk assessment and mitigation	SO2 / SO4	Chapter 3, Section 3.7
Security considerations	Relevant SO	Chapter 3, Section 3.7
Integrated / system-level engineering	SO1 / SO2	Chapter 3, Section 3.5
Project planning and management	SO5	Chapter 4, Section 4.8
Individual contribution	SO1–SO7	Contribution Statement + Chapter 4, Section 4.8